AD23431 - STATISTICAL ANALYSIS AND COMPUTING

EXP 9: R methods for Train and Test data split

Select a dataset of your choice (either built-in datasets like iris, mtcars, or any real-world dataset).

Perform the following tasks:

1. Data Preparation

- Load the dataset into R
- Display the structure and summary of the dataset

PROGRAM:

```
data(iris)
head(iris)
str(iris)
summary(iris)
```

OUTPUT:

```
'data.frame':	150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

2. Train–Test Split using Three Methods

Split the dataset into 70% training data and 30% testing data using the following approaches:

a) Base R Method

PROGRAM:

```
set.seed(123)
index_base <- sample(1:nrow(iris), 0.7*nrow(iris))
train_base <- iris[index_base, ]
test_base  <- iris[-index_base, ]
cat("\n--- Base R Method ---\n")
```

```
print(dim(train_base))
print(dim(test_base))
print(table(train_base$Species))
print(table(test_base$Species))
```

OUTPUT:

```
--- Base R Method ---
[1] 105   5
[1] 45  5

    setosa versicolor  virginica
        36         32         37

    setosa versicolor  virginica
        14         18         13
```

b) Using dplyr Package

PROGRAM:

```
library(dplyr)
set.seed(123)
iris2 <-iris %>% mutate(id = row_number())
train_dplyr<-iris2 %>% sample_frac(0.7)
test_dplyr<-iris2 %>% filter(!id %in% train_dplyr$id)
train_dplyr<-train_dplyr %>% select(-id)
test_dplyr<-test_dplyr %>% select(-id)
cat("\n--- dplyr Method ---\n")
print(dim(train_dplyr))
print(dim(test_dplyr))
print(table(train_dplyr$Species))
print(table(test_dplyr$Species))
```

OUTPUT:

```
--- dplyr Method ---
[1] 105   5
[1] 45  5

    setosa versicolor  virginica
        36         32         37

    setosa versicolor  virginica
        14         18         13
```

c) Using caTools Package

PROGRAM:

```
library(caTools)
set.seed(123)
split<-sample.split(iris$Species, SplitRatio = 0.7)
train_cato<-subset(iris, split == TRUE)
test_cato<-subset(iris, split == FALSE)
cat("\n--- caTools Method ---\n")
print(dim(train_cato))
print(dim(test_cato))
print(table(train_cato$Species))
print(table(test_cato$Species))
```

OUTPUT:

```
--- caTools Method ---
[1] 105   5
[1] 45  5

    setosa versicolor  virginica
        35         35         35

    setosa versicolor  virginica
        15         15         15
```

3. Verification

- Display the dimensions of training and testing datasets for all three methods
- Compare the distribution of the target variable in train and test sets

PROGRAM:

```
cat("\n--- Dimensions ---\n")
cat("\nBase R:\n")
print(dim(train_base))
print(dim(test_base))
cat("\ndplyr:\n")
print(dim(train_dplyr))
print(dim(test_dplyr))
cat("\ncaTools:\n")
```

```
print(dim(train_cato))
print(dim(test_cato))
cat("\n--- Target Variable Distribution ---\n")
cat("\nBase R:\n")
print(table(train_base$Species))
print(table(test_base$Species))
cat("\ndplyr:\n")
print(table(train_dplyr$Species))
print(table(test_dplyr$Species))
cat("\ncaTools:\n")
print(table(train_cato$Species))
print(table(test_cato$Species))
```

OUTPUT:

```
--- Dimensions ---

Base R:
[1] 105   5
[1] 45  5

dplyr:
[1] 105   5
[1] 45  5

caTools:
[1] 105   5
[1] 45  5

--- Target Variable Distribution ---

Base R:

    setosa versicolor  virginica
        36         32         37

    setosa versicolor  virginica
        14         18         13

dplyr:

    setosa versicolor  virginica
        36         32         37
```

```
setosa versicolor  virginica
    14         18         13
```

caTools:

```
setosa versicolor  virginica
    35         35         35

setosa versicolor  virginica
    15         15         15
```